

NORRKOPING, Sweden

WMO: 025740

Lat: 58.583N

Lon: 16.147E

Elev: 136

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.0	10.2	0.3	5.6	4.5	5.9	7.5	12.6	16.4	37.7	14.5	37.9	1.6	260	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	81.1	64.8	77.9	63.8	74.6	62.3	67.7	75.9	66.0	73.6	64.2	71.9	5.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.2	93.7	70.3	63.2	87.1	68.9	61.1	80.8	67.8	32.1	76.2	30.8	73.4	29.4	72.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.5	12.6	11.3	0.7	85.8	8.3	3.4	-5.3	88.3	-10.1	90.2	-14.8	92.2	-20.8	94.6		
(5)			-0.2	70.1	8.0	2.3	-6.0	71.8	-10.7	73.1	-15.2	74.5	-21.0	76.2	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	28.9	29.1	34.4	43.4	52.5	59.3	64.5	62.4	55.2	45.2	38.0	32.6	(6)	(7)
	DBStd	14.18	8.70	8.30	7.83	6.04	5.99	5.08	4.89	4.54	5.32	6.33	6.59	8.80	(7)	(8)
	HDD50	3072	653	587	484	213	41	2	0	0	18	171	362	541	(8)	(9)
	HDD65	7192	1118	1007	948	649	390	181	70	104	296	613	810	1006	(9)	(10)
	CDD50	1447	0	0	1	14	117	282	450	386	172	24	1	0	(10)	(11)
	CDD65	93	0	0	0	0	1	11	55	25	1	0	0	0	(11)	(12)
	CDH74	812	0	0	0	1	31	141	467	168	4	0	0	0	(12)	(13)
	CDH80	143	0	0	0	0	1	21	97	24	0	0	0	0	(13)	(14)
	WSAvg	5.3	5.4	5.4	5.7	5.5	5.3	5.2	4.7	4.6	4.9	5.1	5.5	5.8	(14)	(15)
	PrecAvg	24.1	1.7	1.4	1.2	1.4	1.7	2.7	2.9	2.8	2.3	2.2	2.0	1.9	(15)	(16)
(16) Precipitation	PrecMax	31.3	3.4	2.4	2.2	3.5	3.5	4.9	5.6	6.9	5.1	5.5	4.6	3.5	(16)	(17)
	PrecMin	17.2	0.3	0.6	0.3	0.1	0.3	0.5	0.2	0.4	0.4	0.1	0.4	0.5	(17)	(18)
	PrecStd	2.9	0.8	0.5	0.6	0.9	0.8	1.1	1.3	1.3	1.3	1.1	1.0	0.8	(18)	(19)
															(19)	(20)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.4	48.1	58.0	68.9	78.0	83.1	86.7	82.5	74.2	62.9	54.7	52.2	(20)	(21)
		MCWB	44.4	43.2	47.3	53.0	60.9	64.1	66.8	65.4	62.0	56.5	52.0	48.6	(21)	(22)
	2%	DB	44.3	44.3	52.7	63.4	73.3	78.3	82.6	78.3	70.6	59.2	51.5	48.2	(22)	(23)
		MCWB	41.7	40.8	44.3	50.5	58.0	62.5	65.5	65.0	60.3	54.8	49.1	45.3	(23)	(24)
	5%	DB	41.9	41.9	48.5	58.9	68.9	74.1	79.2	74.9	67.0	56.7	49.3	45.5	(24)	(25)
		MCWB	39.5	39.1	42.2	48.4	56.1	60.1	64.6	63.6	58.6	53.1	47.1	43.0	(25)	(26)
	10%	DB	39.6	39.5	45.2	54.8	64.8	70.8	75.8	71.9	64.2	54.5	47.0	43.4	(26)	(27)
		MCWB	37.5	36.9	40.4	45.6	53.8	58.6	63.7	61.9	57.5	51.3	45.1	41.2	(27)	(28)
	0.4%	WB	44.9	43.9	48.4	55.5	63.3	66.8	70.5	70.0	64.1	58.6	52.5	48.8	(28)	(29)
		MCDB	46.9	46.7	56.3	66.6	73.5	76.9	79.1	77.5	70.3	61.7	54.2	51.5	(29)	(30)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.1	41.3	45.0	51.9	60.3	64.1	68.2	67.3	61.5	55.7	49.3	45.6	(30)	(31)
		MCDB	44.0	43.7	51.1	60.7	69.2	75.0	77.1	73.9	67.9	58.3	51.0	47.7	(31)	(32)
	5%	WB	39.7	39.4	43.0	49.2	57.6	61.9	66.8	65.4	59.8	53.6	47.2	43.2	(32)	(33)
		MCDB	41.6	41.7	47.6	57.6	66.0	71.3	75.1	71.6	65.4	56.1	48.9	45.1	(33)	(34)
	10%	WB	37.7	37.2	40.8	46.7	55.2	59.9	65.1	63.6	58.2	51.7	45.3	41.1	(34)	(35)
		MCDB	39.2	39.2	44.5	53.3	63.1	68.4	73.2	69.7	62.9	54.3	46.8	42.9	(35)	(36)
	5% DB	MDBR	8.2	9.6	13.3	17.2	18.1	17.0	16.5	15.8	14.3	11.5	7.8	7.9	(36)	(37)
		MCDBR	8.8	10.3	18.0	25.1	25.8	24.5	22.9	20.7	18.8	13.5	9.7	8.7	(37)	(38)
		MCWBR	8.2	7.8	11.3	14.0	12.9	11.0	8.8	9.3	10.2	9.5	8.3	8.1	(38)	(39)
		MCWBR	8.7	9.4	16.4	22.1	21.7	21.0	18.6	17.3	17.5	12.3	9.5	8.5	(39)	(40)
(40) Clear-Sky Solar Irradiance	taub		0.284	0.321	0.340	0.378	0.374	0.362	0.389	0.380	0.348	0.338	0.310	0.270	(40)	(41)
		taud	2.243	2.297	2.351	2.309	2.386	2.441	2.387	2.409	2.484	2.419	2.290	2.177	(41)	(42)
	Ebn at Noon		173	222	255	262	272	277	266	259	252	215	160	135	(42)	(43)
		Edh at Noon	15	22	28	34	34	33	34	31	25	20	14	11	(43)	(44)
(44) All-Sky Solar Radiation	RadAvg		107	295	772	1263	1663	1842	1672	1330	893	401	138	74	(44)	(45)
	RadStd		15	50	80	130	136	85	166	128	111	62	29	9	(45)	(46)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (56 neighbors)		+0.70	N/A	N/A	N/A	N/A	N/A	-232	-284	N/A	N/A	

Nomenclature: See separate page